Class-ECS-II Div-C

## UNIT-I Introduction to Data Communication and Networking:

Data Communication are the exchange of data between two or more devices via some form of transmission medium such as a wire cable wireless. For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs).

### Characteristics of a data communications system

The effectiveness of a data communications system depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter.

### 1-Delivery:

The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.

**Meaning:** Data goes to the right place.

**Example:** Your message should go to *your friend*, not someone else.

### 2-Accuracy:

The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.

**Meaning:** Data must not change.

**Example:** If you send “Hello,” it should not arrive as “H3llo.”

### 3-Timeliness:

The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called *real-time* transmission.

**Meaning:** Data must arrive on time.

**Example:** A video call must show what’s happening *right now*, not a few seconds later.

### 4-Jitter:

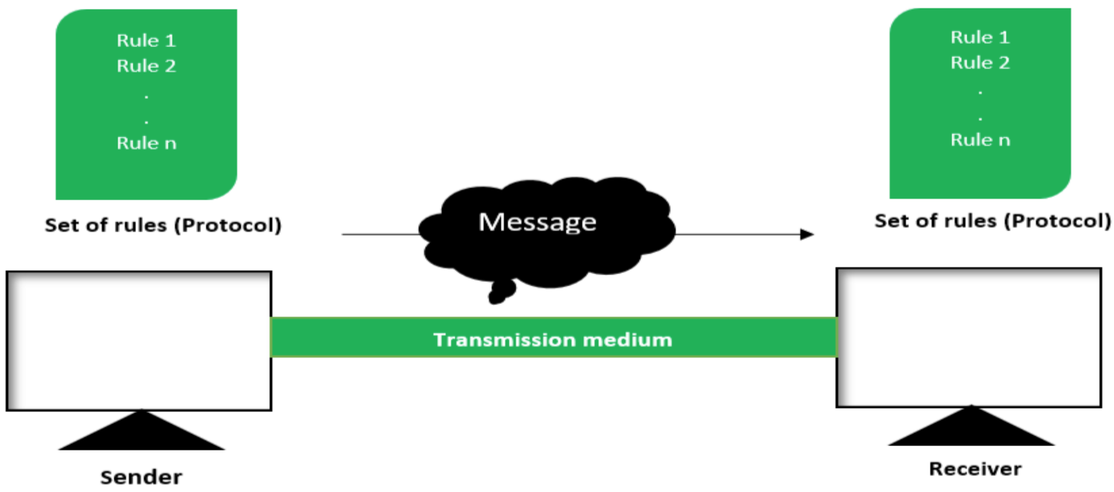
Jitter refers to the variation in the packet arrival time. It is the uneven (sometimes fast, sometimes slow) delay in the delivery of audio or video packets. For example, let us assume that video packets are sent every 3D ms. If some of the packets arrive with 3D-ms delay and others with 4D-ms delay, an uneven quality in the video is the result.

**Meaning:** Data arrives at uneven times.

**Example:** In a video call, some frames come late, making the video look jumpy or shaky.

## Components

A data communications system has five components.




## 1. Sender

The sender is the **device that starts the communication**. It creates and sends the data. The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.

### Example:

When you send a WhatsApp message from your phone, **your phone is the sender**.

2. Receiver

The receiver is the **device that gets the data** from the sender. The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.

Example:
Your friend’s phone that receives your WhatsApp message is the **receiver**.

3. Message

The message is the **actual information** being sent. The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.

Examples:

- A text message: “Hello!”
- A photo you share
- A voice message
- A video file
- These are all **messages** in data communication.

4. Transmission Medium (Communication Channel)

The medium is the **path through which the data travels** from the sender to the receiver. The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.

Types of Mediums:

1. **Wired** → copper cables, fiber-optic cables
2. **Wireless** → Wi-Fi, Bluetooth, radio waves

Example:

- When using **Wi-Fi**, messages travel through the **air using radio waves**.
- When using **Ethernet cable**, data travels through the **cable**.

5. Protocol

Protocol means the **set of rules** that both the sender and receiver follow so they can communicate correctly. A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating.

Example:

WhatsApp uses protocols that decide:

- how messages are formatted
- how messages are encrypted
- how errors are handled

Other common protocols:

- **HTTP** → for browsing websites
- **TCP/IP** → for sending data on the internet
- **SMTP** → for sending email

For example,

**Sender:** Your **smartphone** sends the WhatsApp message.
**Message:** The **text, image, or voice note** you are sending.
**Medium:** The message travels through **mobile data (4G/5G) or Wi-Fi → Internet servers → Friend's network**.
**Protocol: TCP/IP** ensures reliable delivery, and **WhatsApp's messaging protocol** handles message formatting and encryption.
**Receiver:** Your **friend's smartphone** receives and displays the message.

Data Representation:

Data representation refers to how different kinds of data are stored, processed, and transmitted in a computer or network.

Computers use binary digits (0s and 1s), so all forms of data must ultimately be converted into binary form.

Types of Data Representation

1. Text

- Text is represented using character encoding schemes.
- Each character (letter, digit, symbol) is assigned a numeric value.

Common Encoding Methods

- ASCII (American Standard Code for Information Interchange)
  - Uses 7 bits (128 characters).
- Extended ASCII – 8 bits (256 characters)
- Unicode (UTF-8, UTF-16)
  - Supports all world languages and special symbols.

**Example**

- Character ‘A’ → ASCII code 65 → Binary: 01000001

**2. Numbers**

- Numbers are already in numeric form, so they are directly represented in binary number system.
- Computers can store:
  - Integers (whole numbers)
  - Floating-point numbers (decimals)

**Examples**

- Decimal 10 → Binary: 1010
- Decimal 25.5 → Represented using floating-point format (IEEE 754)

**3. Images**

- Images are made of tiny dots called pixels.
- Each pixel has:
  - Color information (Red, Green, Blue values)
  - Brightness

Image Representation

- Bitmap (BMP): Stores every pixel value.
- JPEG / PNG: Use compression to reduce file size.

**Example**

- A pixel color might be stored as:
  - R = 255
  - G = 100
  - B = 0

Combined binary values represent the color.

**4. Audio**

- Audio is represented as digital samples of sound waves.
- The sound wave is sampled at regular intervals (Sampling Rate).

Key Concepts

- Sampling Rate (e.g., 44.1 kHz for CD quality)

- Bit Depth (e.g., 16-bit audio)

How Audio is Stored

- More samples → better quality
- Higher bit depth → more accurate sound representation

5. Video

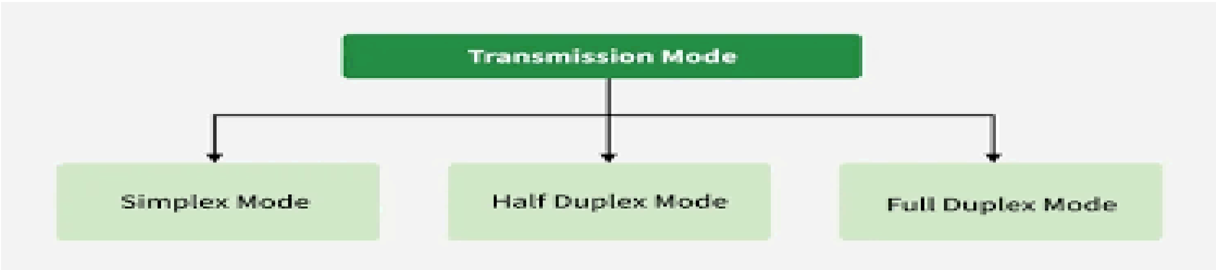
- Video = Sequence of images (frames) + Audio
- Each frame is stored like a digital image.
- Played quickly (e.g., 30 frames per second) to create motion.

Video Representation

- Frame resolution (e.g., 1920 × 1080)
- Frame rate (30 fps / 60 fps)
- Compression (MP4, AVI, MKV)

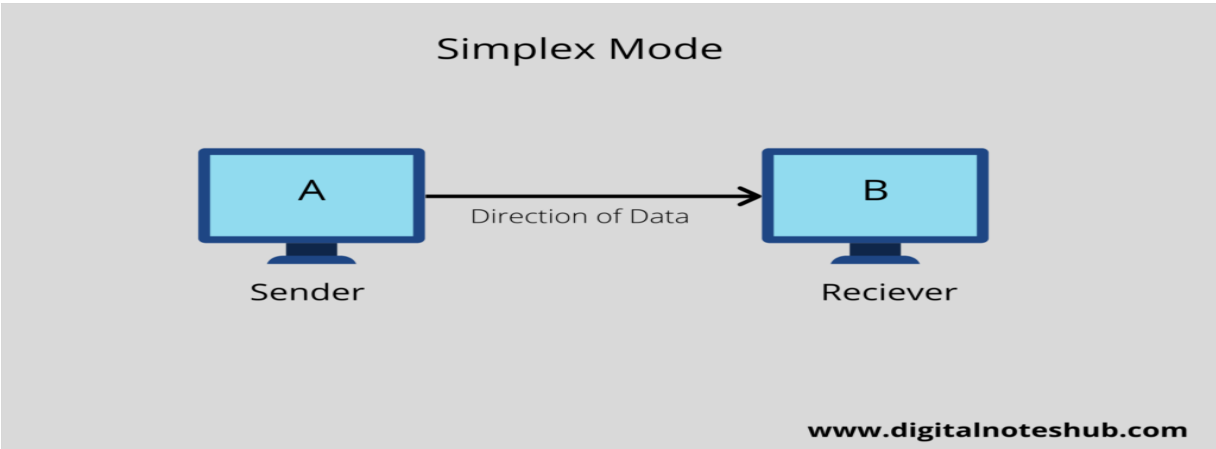
Data Flow

Communication between two devices can be simplex, half-duplex, or full-duplex.




Simplex

In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive).



Keyboards and traditional monitors are examples of simplex devices. The keyboard can only introduce input; the monitor can only accept output. The simplex mode can use the entire capacity of the channel to send data in one direction.

**Meaning:**

Data flows in only one direction.

One device only sends and the other only receives.

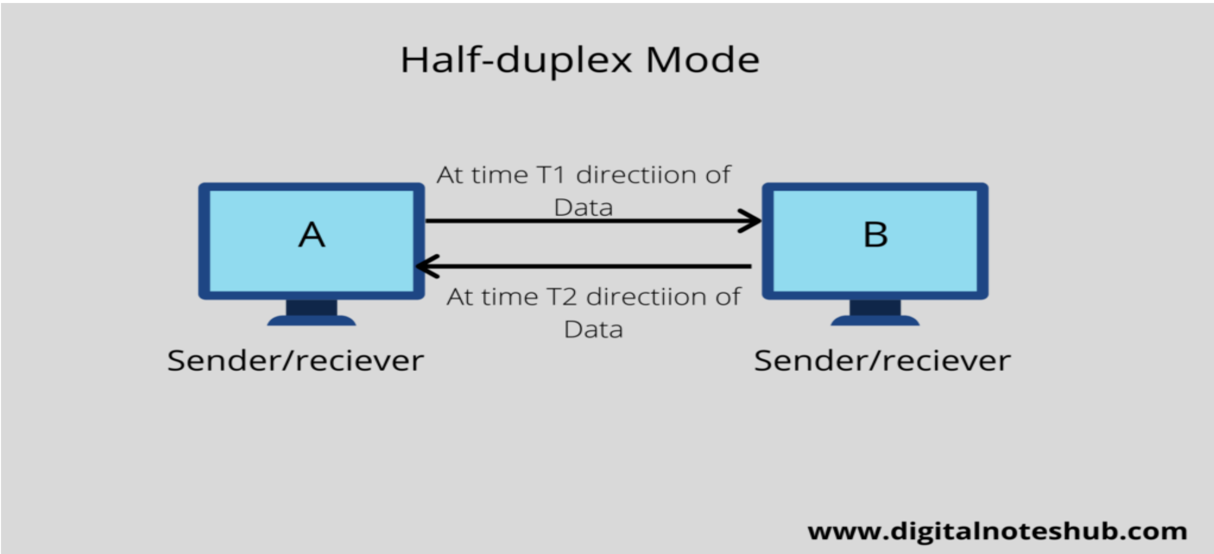
**Example:**

- Keyboard → computer (keyboard only sends input)
- TV broadcast (TV receives signals but doesn't send anything back

### Half-Duplex

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time. Walkie-talkies and CB (citizens band) radios are both half-duplex systems.

The half-duplex mode is used in cases where there is no need for communication in both directions at the same time; the entire capacity of the channel can be utilized for each direction.




**Meaning:**

Data flows in both directions, but not at the same time.

Devices take turns sending and receiving.

**Example:**

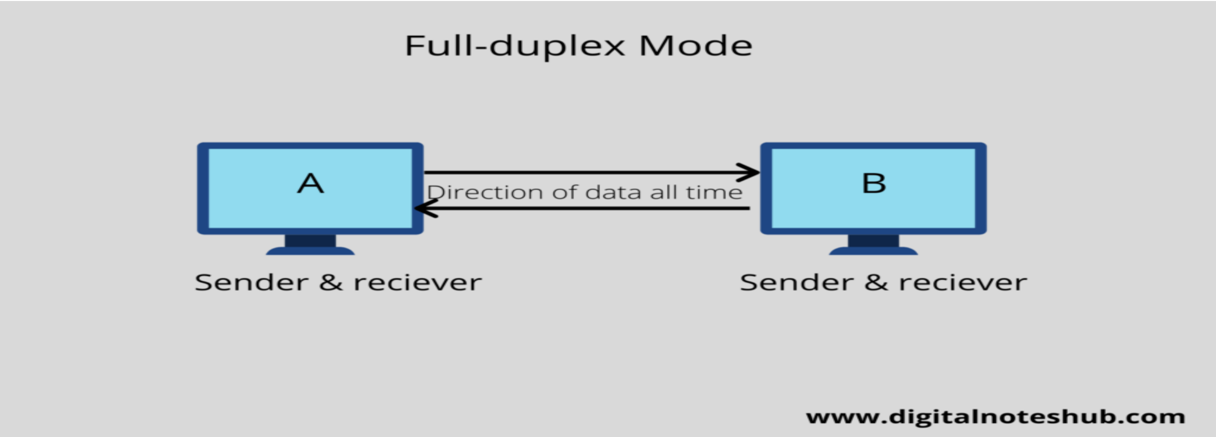
- Walkie-talkie
   - When one person talks, the other must wait.
   - They cannot talk at the same time.

**Full-Duplex**

In full-duplex mode, both stations can transmit and receive simultaneously in full-duplex mode, signals going in one direction share the capacity of the link: with signals going in the other direction. This sharing can occur in two ways: Either the link must contain two physically separate transmission paths, one for sending and the other for receiving; or the capacity of the channel is divided between signals traveling in both directions.

One common example of full-duplex communication is the telephone network. When two people are communicating by a telephone line, both can talk and listen at the same time.

The full-duplex mode is used when communication in both directions is required all the time. The capacity of the channel must be divided between the two directions.




**Meaning:**

Data flows in both directions at the same time.

Devices can send and receive simultaneously.

**Example:**

- Phone call
   - Both people can talk and listen at the same time.
- Internet browsing

- Your device sends requests while receiving data at the same time.

Data Flow Communication Model Computer N/W:
A communication model explains how data travels from one place to another.
It shows all the basic components involved in sending and receiving information.

Source → Transmitter → Transmission System → Receiver → Destination

1. Source (Sender)

- The place where the message starts.
- Example: A person speaking or a computer sending data.

2. Transmitter

- Changes the message into a signal so it can travel.
- Example: A modem or a network card.

3. Transmission System

- The path through which the signal travels.
- Example: Cable, Wi-Fi, fiber, radio waves.

4. Receiver

- Gets the signal and changes it back into the original message.
- Example: A modem or phone receiver.

5. Destination

- The final place where the message is delivered.
- Example: The person hearing the message or a computer receiving data.

## Difference Between Simplex, Half-Duplex & Full-Duplex:-

| Feature | Simplex | Half-Duplex | Full-Duplex |
|---|---|---|---|
| Direction of Data | One-way only | Two-way, but one direction at a time | Two-way simultaneously |
| Communication | Unidirectional | Bidirectional (alternating) | Bidirectional at the same time |
| Example | TV broadcast, Keyboard → Computer | Walkie-talkie, Old Ethernet | Phone call, Video call, Modern Ethernet |
| Speed | Low | Moderate | High |
| Complexity | Simple | Moderate | Complex |
| Feedback | Not possible | Possible but not simultaneous | Possible simultaneously |
| Hardware Requirement | Minimal | Moderate | More advanced hardware required |
| Efficiency | Less efficient | More efficient than simplex | Most efficient |
| Cost | Low | Moderate | High |
| Usage | Simple data transfer | Networks where turn-taking is okay | Real-time communication like calls, streaming |

## Introduction to Computer Networks:

A **computer network** is a system in which multiple computers or devices are connected together to share resources, exchange data, and communicate with each other. These networks enable communication across small local spaces (like homes or offices) or across the world (like the internet).

### Why computer networks are important

- Enable sharing of files, data, and applications
- Allow communication via email, video calls, messaging
- Support resource sharing (printers, storage, internet access)
- Improve efficiency and collaboration
- Enable access to remote services and cloud computing.

### What is a Computer Network?

A computer network is a group of two or more computers or devices that are connected together so they can share information, resources, and services.

**In simple words:**

A network is a connection of devices that can talk to each other.

Examples:

- Two computers connected with a cable
- A school computer lab
- The internet (largest network in the world)

### Why Do We Need Networks?

We need networks because they make communication and sharing easier, faster, and cheaper.

Main reasons:

a) Communication

- Send emails
- Chat, video calls
- Share messages and files instantly

b) Sharing Resources

- Share printers, scanners, storage
- No need to buy separate devices for each user

c) Sharing Data and Information

- Access common files

- Share documents within offices or schools

d) Centralized Management

- Easy to manage users, security, and data from a single place

e) Cost Saving

- No need for multiple hardware devices
- Less maintenance cost

f) Internet Access

- All connected devices can access the internet through the network

Basic Idea: Connecting Two or More Devices

A network works by connecting devices using:

- Cables (wired network)
- Wi-Fi (wireless network)
- Optical fiber

When devices are connected, they can send and receive data.
Each device in the network can act as:

- Sender → sends data
- Receiver → receives data

Example:

- Your mobile phone connects to Wi-Fi → you can browse the internet
- A computer sends a file to a printer → printer prints the file

Computer Network Architecture:

Computer Network Architecture is defined as the physical and logical design of the software, hardware, protocols, and media of the transmission of data. Simply we can say that how computers are organized and how tasks are allocated to the computer. The two types of networks architecture

1. Peer-To-Peer network
2. Client/Server network

1) Peer-To-Peer network:

Peer-To-Peer network is a network in which all the computers are linked together with equal privilege and responsibilities for processing the data.

Peer-To-Peer network is useful for small environments, usually up to 10 computers. Peer-To-Peer network has no dedicated server. Special permissions are assigned to each computer for sharing the resources, but this can lead to a problem if the computer with the resource is down.

### Advantages of Peer-To-Peer Network:

It is less costly as it does not contain any dedicated server. If one computer stops working but, other computers will not stop working. o It is easy to set up and maintain as each computer manages itself.

### Disadvantages of Peer-To-Peer Network:

In the case of Peer-To-Peer network, it does not cannot back up the data as the data is different in different locations. It has a security issue as the device is managed itself.

### 2)Client/Server Network:

Client/Server network is a network model designed for the end users called clients, to access the resources such as songs, video, etc. from a central computer known as Server.

The central controller is known as a called client. o A server performs all the major operations such as security and network management. o A server is responsible for managing all the resources such as files, directories, printer, etc. o All the clients communicate with send some data to client 2, then it first sends the request to the server for the permission. The server sends the response to the client 1 to initiate its communication with the client 2.

### Advantages of Client/Server network:

1. A Client/Server network contains the centralized system. Therefore, we can back up the data easily. A Client/Server network has a dedicated server that improves the overall performance of the whole system.
2. Security is better in Client/Server network as a single server administers the shared resources.
3. It also increases the speed of the sharing resources.

**Disadvantages of Client/Server network:**

1. Client/Server network is expensive as it requires the server with large memory.
2. A server has a Network Operating System (NOS) to provide the resources to the clients, but the cost of NOS is very high.
3. It requires a dedicated network administrator to manage all the resources

**Components of Computer Networks:**

Computer networks consist of **hardware**, **software**, and **protocols** that work together.

A computer network consists of:

1. Hardware – Router, switch, modem, NIC, AP
2. Transmission Media – Cables (wired), radio/microwave (wireless)
3. Software – NOS, protocols, network management tools
4. Protocols – TCP/IP, HTTP, DNS, DHCP, etc.
5. Users/Clients – End devices
6. Servers – Provide services and resources

**NIC (Network Interface Card)** in a computer network is a **hardware component** that allows a computer or device to **connect to a network** and communicate with other devices.

**What a NIC does**

- Provides a **physical interface** between a computer and a network
- Converts data from the computer into signals that can travel over a network (and vice versa)
- Enables communication over **LAN, WAN, or the Internet**

**Key features of a NIC**

- **MAC Address**:

Every NIC has a unique **Media Access Control (MAC) address** used to identify the device on a network.

- **Speed**:
  Common speeds include 10 Mbps, 100 Mbps, 1 Gbps, 10 Gbps.
- **Duplex mode**:
  Supports half-duplex or full-duplex communication.

**Types of NIC**

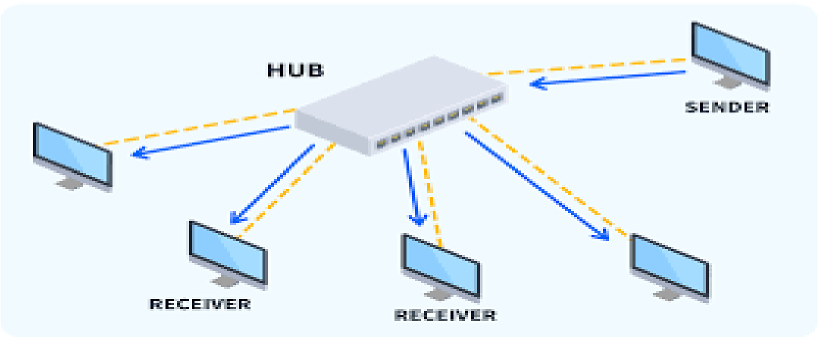
1. **Wired NIC**
   - Uses Ethernet cables (RJ-45)

- More stable and faster

2. **Wireless NIC**
   - Uses Wi-Fi (radio waves)
   - Allows mobility

### 1-Hub

- A basic network device that broadcasts data to all connected devices.
- Causes data collisions → less secure and slower.
- Used rarely today, replaced by switches.
- **Use case:** Small networks or simple office networks.
- **Limitation:** Inefficient for large networks because it can cause **data collisions**.
- Types of Hubs:
- **Active Hub:** Boosts signals, acts as a repeater.
- **Passive Hub:** Relays signals without amplification.
- **Intelligent Hub:** Provides monitoring and management functions.




### 2-Switch

- Connects devices within the same LAN.
- Uses MAC addresses to forward data only to the intended device.
- Improves speed, security, and reduces network collisions.

### Switch in Computer Networks (Detailed Explanation)

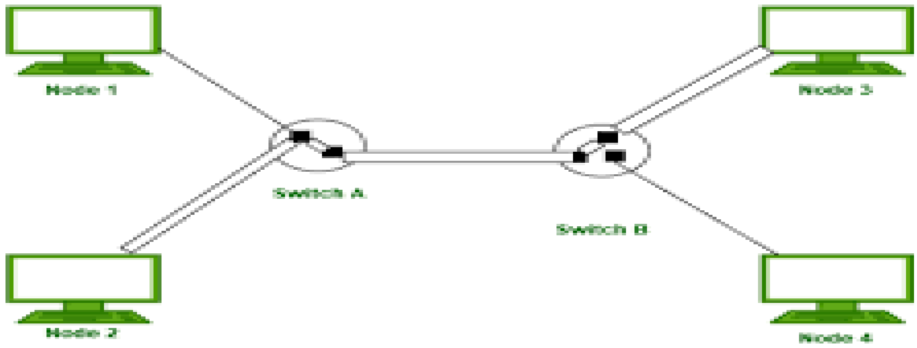
A **network switch** is a **Layer 2 (Data Link Layer)** device used in computer networks to connect multiple devices (computers, printers, servers, etc.) within a **Local Area Network (LAN)** and efficiently forward data between them.

### Definition

A **switch** is a networking device that receives data frames and forwards them **only to the intended destination device** using **MAC addresses**, reducing unnecessary network traffic.

How a Switch Works

1. When a device sends data, the switch receives the **Ethernet frame**.
2. The switch reads the **source MAC address** and stores it in the **MAC address table (CAM table)** along with the port number.
3. The switch checks the **destination MAC address**:
    - If found → forwards the frame to the specific port.
    - If not found → floods the frame to all ports (except the source port).
4. Over time, the switch builds a complete MAC address table and works efficiently.




- **Types:**
    - Unmanaged Switch (plug-and-play)
    - Managed Switch (configurable)

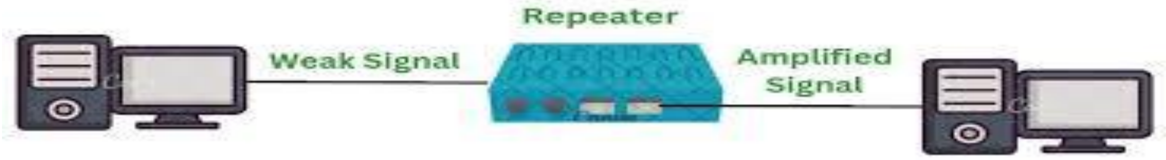
**Advantages:**

- Reduces collisions.
- Increases network efficiency.
- Can connect more devices than a hub.

3-**Repeater**

- Amplifies and regenerates weak signals to extend network distance.
- Used in long cable segments or wireless networks.
- Function: **Extends the reach of a network.**
- How it works: Amplifies or regenerates weak signals **so data can travel longer distances without degradation.**
- Use case: **Long office buildings or campus networks.**

A repeater is a network device used to extend the distance of a network by regenerating and amplifying signals that become weak during transmission.




Definition

A repeater is a Layer 1 (Physical Layer) device that receives a weak or distorted signal, cleans it, and retransmits it at its original strength.

Why Repeater is Needed

- Signals weaken due to attenuation
- Noise and interference affect data transmission
- Long-distance cables reduce signal quality

How a Repeater Works

1. Receives weak signal
2. Removes noise and distortion
3. Regenerates the signal
4. Sends the refreshed signal forward

It does not understand data, MAC address, or IP address.

Types of Repeaters

1. Analog Repeater

- Amplifies analog signals
- Used in traditional communication systems

2. Digital Repeater

- Regenerates digital signals
- Used in computer networks

3. Wireless Repeater

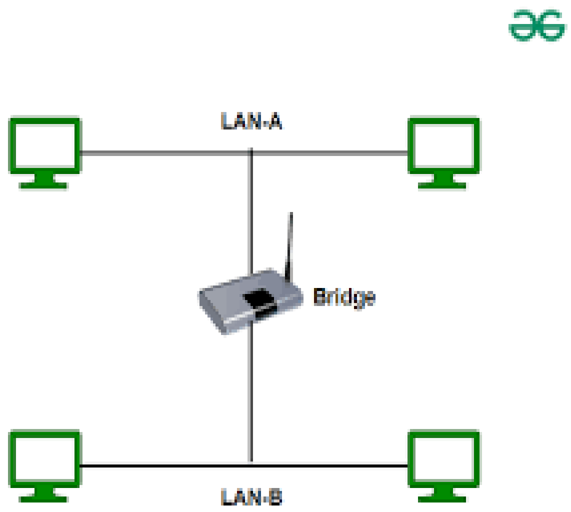
- Extends Wi-Fi coverage
- Also called Wi-Fi extende

## Bridge

- Connects two LAN segments and filters traffic using MAC addresses.
- Reduces network congestion.
- Operates at the Data Link Layer (Layer 2).
- Connects and filters traffic between two LAN segments.
- Uses MAC addresses to forward data only to the intended segment.
- Types of Bridges:
- **Transparent Bridge:** Learns MAC addresses automatically.
- **Source Routing Bridge:** Follows routes specified by the sender.

### How a Bridge Works

1. A bridge receives a data frame.
2. It reads the **destination MAC address**.
3. The bridge checks its **bridge table**:
   - If the destination is on the same segment → frame is **discarded**
   - If the destination is on another segment → frame is **forwarded**
   - If unknown → frame is **broadcast**
4. The bridge learns MAC addresses dynamically.




## Gateway

- Connects networks that use different protocols.
- Performs protocol translation (e.g., VoIP gateway).
- Acts as a protocol converter.
- Connects two networks using different architectures or protocols.
- Can work at any OSI layer depending on its function.

- **Examples:** connecting an enterprise LAN to the internet.


Router

- A router connects multiple networks (e.g., home network to the internet).
- It forwards data packets based on IP addresses.
- **Performs:**
  - Routing (choosing best path for data)
  - Network Address Translation (NAT)
  - DHCP (assigning IP addresses to devices)
- Usually used in homes, offices, and ISPs.
- Operates at the Network Layer (Layer 3).

Protocol Software

Implements communication protocols (TCP/IP, HTTP, FTP, etc.).

Network Protocols

Protocols are rules that define how data moves in a network.

Key Protocols:

- TCP/IP – Core of internet communication
- HTTP/HTTPS – Web browsing
- FTP/SFTP – File transfer
- SMTP/POP/IMAP – Email communication
- DNS – Converts domain names to IP addresses
- DHCP – Automatically assigns IP addresses

## Network Topology

**A network topology** is the arrangement of devices (nodes) and connections (links) in a computer network. It shows how computers, servers, and other devices are connected and how data flows between them. There are two main types of **topologies:**

- Physical Topology: The actual physical layout of cables and devices.
- Logical Topology: How data moves across the network, regardless of physical layout.

*Note: Choosing the right topology is important because it affects the performance, cost, reliability, and security of the network.*

**Bus** → straight line
**Star** → center with spokes
**Ring** → circle
**Mesh** → many lines
**Tree** → branches
**Hybrid** → mix of shapes

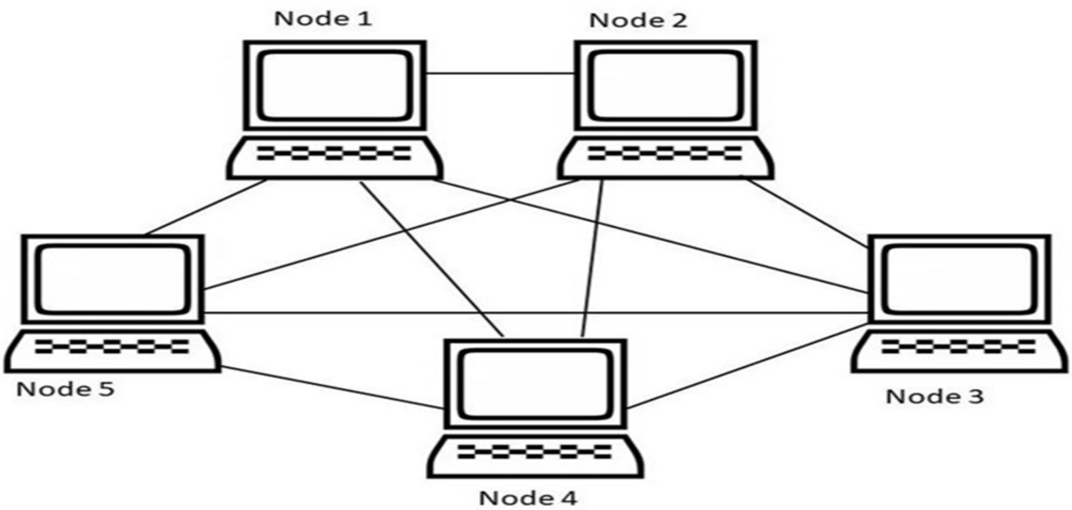
## Why is Network Topology Important?

Network Topology is important because it defines how devices are connected and how they communicate in the network. Here are some points that defines why network topology is important.

- **Network Performance:** Upon choosing the appropriate topology as per requirement, it helps in running the network easily and hence increases network performance.
- **Network Reliability:** Some topologies like Star, Mesh are reliable as if one connection fails, they provide an alternative for that connection, hence it works as a backup.
- **Network Expansion**: Choosing correct topology helps in easier expansion of Network as it helps in adding more devices to the network without disrupting the actual network.
- **Network Security:** Network Topology helps in understanding how devices are connected and hence provides a better security to the network.

## 1- Mesh Topology

In a mesh topology, every device is connected to another device via a particular channel. Every device is connected to another via dedicated channels. These channels are known as links. In Mesh Topology, the protocols used are AHCP (Ad Hoc Configuration Protocols), DHCP (Dynamic Host Configuration Protocol), etc.




- Suppose, the N number of devices are connected with each other in a mesh topology, the total number of ports that are required by each device is $N-1$. In Figure, there are 6 devices connected to each other, hence the total number of ports required by each device is 5. The total number of ports required = N * (N-1).

- Suppose, N number of devices are connected with each other in a mesh topology, then the total number of dedicated links required to connect them is $NC_2$ i.e. N * (N-1)/2. In Figure, there are 6 devices connected to each other, hence the total number of links required is 6 * 5/2 = 15.

### How it works (Flow):

1. When a device wants to send data, it checks available paths.
2. Data is sent through the **shortest or fastest path**.
3. If one link fails, data is rerouted through another link.
4. The data reaches the destination without interruption.

### Advantages:

1. Mesh topology is very reliable because every node is connected to every other node.
2. There is no single point of failure in a full mesh network.
3. Data transfer is fast since each connection is dedicated.
4. It works well under heavy network traffic.

Disadvantages:

1. Mesh topology is very expensive due to the large number of cables and ports required.
2. It is complex to set up because each device must be connected to every other device.
3. It requires a lot of cables, which can be difficult to manage.
4. Maintenance and troubleshooting are difficult due to the complex connections.

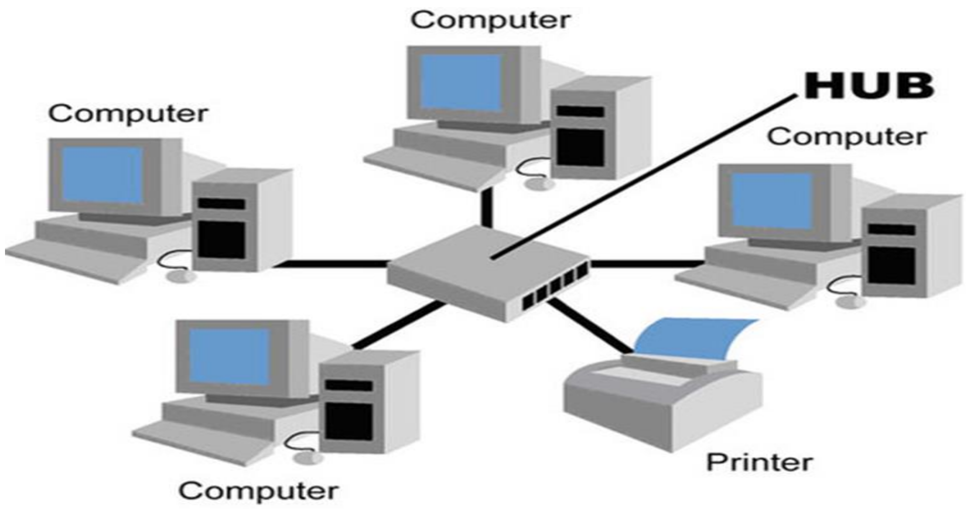
***Note: A common example of mesh topology is the internet backbone, where various internet service providers are connected to each other via dedicated channels. This topology is also used in military communication systems and aircraft navigation systems.***

2-Star Topology

In Star Topology, all the devices are connected to a single hub through a cable. This hub is the central node and all other nodes are connected to the central node. The hub can be passive in nature i.e., not an intelligent hub such as broadcasting devices, at the same time the hub can be intelligent known as an active hub. Active hubs have repeaters in them.

Star Topology

***Note: Here, Coaxial cables or RJ-45 cables are used to connect the computers & many popular Ethernet LAN protocols are used as CD(Collision Detection), CSMA (Carrier Sense Multiple Access), etc.***




How it works (Flow):

1. The sender transmits data to the hub.
2. The hub receives the data signal.
3. The hub **broadcasts the data to all connected devices**.
4. The intended receiver accepts the data; others ignore it.

Advantages:

1. Star topology is easy to install because all nodes connect to a central hub.
2. It is easy to manage since the hub acts as a central control point.
3. Troubleshooting is simple because problems can be isolated to one node.
4. Failure of one node does not affect the rest of the network.
5. If N devices are connected to each other in a star topology, then the number of cables required to connect them is N. So, it is easy to set up.
6. Each device requires only 1 port i.e. to connect to the hub, therefore the total number of ports required is N.

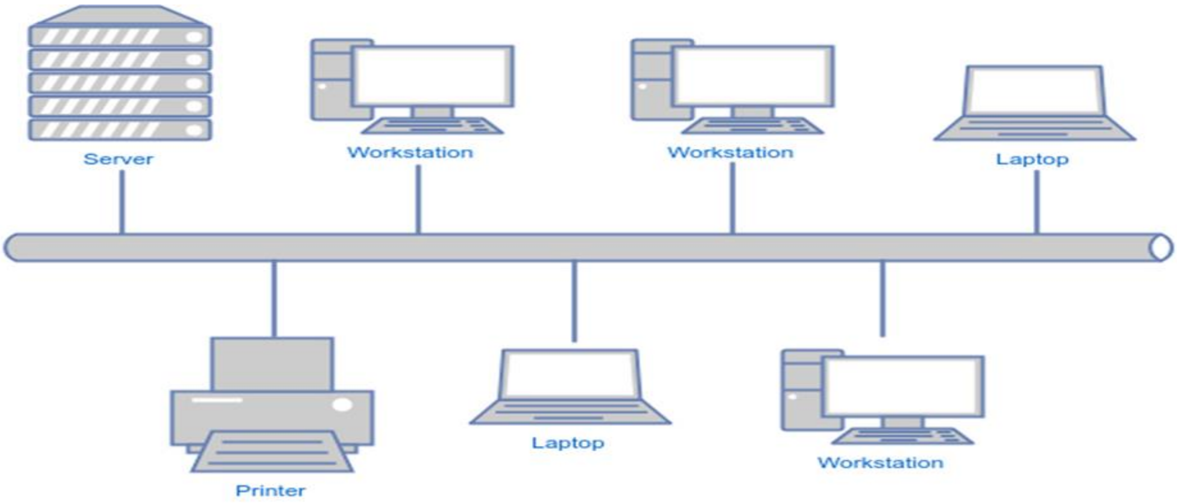
Disadvantages:

1. If the central hub fails, the entire network stops working.
2. Star topology requires more cables than a bus topology.
3. Network performance depends on the hub's capacity.
4. It is more expensive than bus topology.

*Note: A common example of star topology is a local area network (LAN) in an office where all computers are connected to a central hub. This topology is also used in wireless networks where all devices are connected to a wireless access point.*

## 3-Bus Topology

Bus Topology is a network type in which every computer and network device is connected to a single cable. It is bi-directional. It is a multi-point connection and a non-robust topology because if the backbone fails the topology crashes. In Bus Topology, various MAC (Media Access Control) protocols are followed by LAN ethernet connections like TDMA, Pure Aloha, CDMA, Slotted Aloha, etc.




Bus Topology Network

### How it works (Flow):

1. The sender places data on the backbone cable.
2. Data travels in both directions along the cable.
3. All devices listen to the data.
4. The destination device recognizes the address and receives the data.

### Advantages:

1. Bus topology is easy to set up with a simple backbone cable.
2. It uses less cable compared to star or mesh topologies.
3. It is inexpensive to implement.
4. Bus topology works well for small networks with few devices.
5. If N devices are connected to each other in a bus topology, then the number of cables required to connect them is 1, known as backbone cable, and N drop lines are required.
6. Coaxial or twisted pair cables are mainly used in bus-based networks that support up to 10 Mbps.

### Disadvantages:

1. A failure in the main cable stops the entire network.
2. Troubleshooting problems can be difficult.

3. The network slows down as more devices are added.
4. It is not suitable for large networks and is not scalable.

***Note: A common example of bus topology is the Ethernet LAN, where all devices are connected to a single coaxial cable or twisted pair cable. This topology is also used in cable television networks.***

### 4-Ring Topology

In a Ring Topology, it forms a ring connecting devices with exactly two neighboring devices.

Ring topology is a network structure where **each device is connected to two neighboring devices**, forming a closed loop.

**How it works (Flow):**

1. Data is sent in **one direction** around the ring.
2. Each device receives the data and checks the address.
3. If not the destination, the device forwards the data to the next node.
4. The destination device receives the data and stops forwarding.

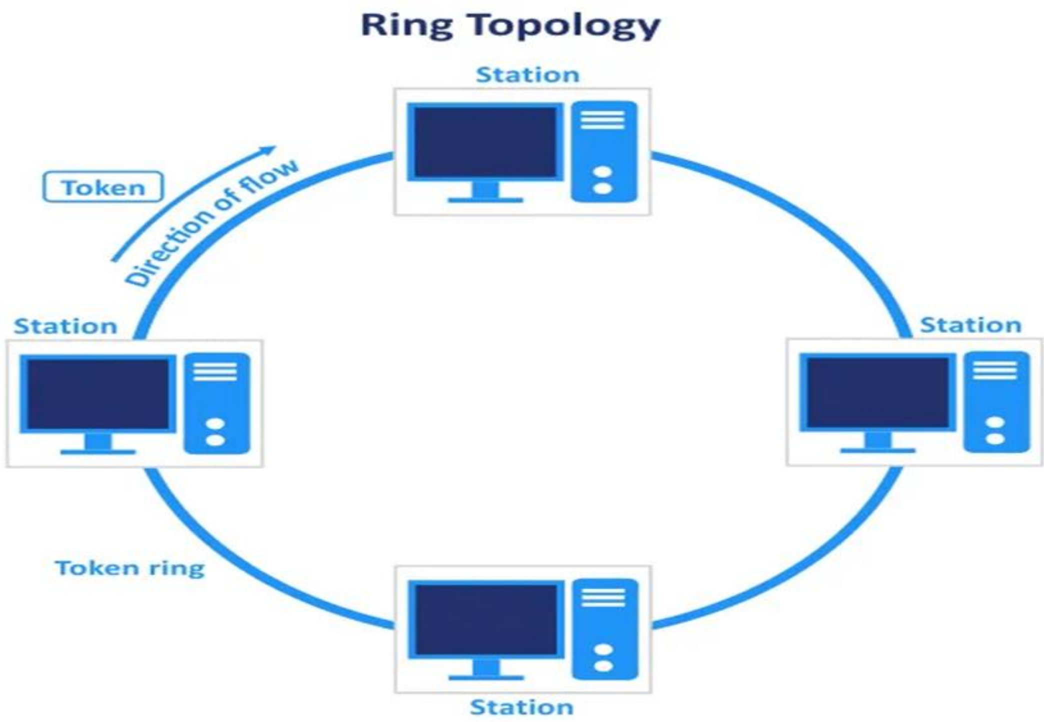
**Advantages:**

1. Ring topology allows data to flow in one direction, reducing collisions.
2. It is easy to install in small networks.
3. All devices have equal access to the network.
4. It provides organized and predictable data transfer.

**Disadvantages:**

1. If one node fails, the entire network can stop functioning.
2. Troubleshooting can be difficult in a ring network.
3. Adding new devices may temporarily disrupt the network.

In-Ring Topology, the Token Ring Passing protocol is used by the workstations to transmit the data where, Token passing is a network access method in which a token is passed from one node to another node & Token is a frame that circulates around the network.




*Note: Here, data flows in one direction, but it can be made bidirectional by having 2 connections between each Network Node, it is called Dual Ring Topology.*

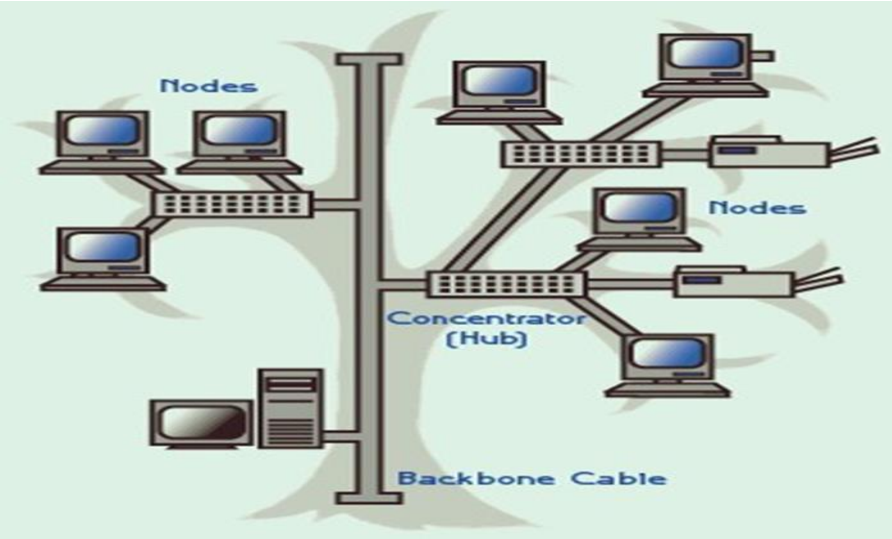
Operations of Ring Topology

- One station is known as a monitor station which takes all the responsibility for performing the operations.
- To transmit the data, the station has to hold the token. After the transmission is done, the token is to be released for other stations to use.
- When no station is transmitting the data, then the token will circulate in the ring.

5-Tree Topology

Tree topology is the variation of the Star topology. This topology has a hierarchical flow of data. In Tree Topology, protocols like DHCP and SAC (Standard Automatic Configuration) are used.

- Here, various secondary hubs are connected to the central hub which contains the repeater.
- This data flow from top to bottom i.e. from the central hub to the secondary and then to the devices or from bottom to top i.e. devices to the secondary hub and then to the central hub.
- It is a multi-point connection and a non-robust topology because if the backbone fails the topology crashes.




*Note: A common example of a tree topology is the hierarchy in a large organization. CEO is the root, who is connected to the different departments (child nodes) of the company, managers overseeing different teams (grandchild nodes) & team members (leaf nodes) are at the bottom of the hierarchy.*

Tree topology is a **hierarchical network structure** formed by connecting multiple star networks to a main backbone cable.

How it works (Flow):

1. Data is sent from a lower-level node to its parent node.
2. Parent nodes forward the data upward or downward in the hierarchy.
3. Data travels through intermediate nodes.
4. The destination node receives the data.

Advantages:

1. Tree topology is easy to expand by adding new branches.
2. It provides a hierarchical structure for better organization.
3. Individual branches can be managed and troubleshooted separately.
4. It works well for large networks like campuses or offices.

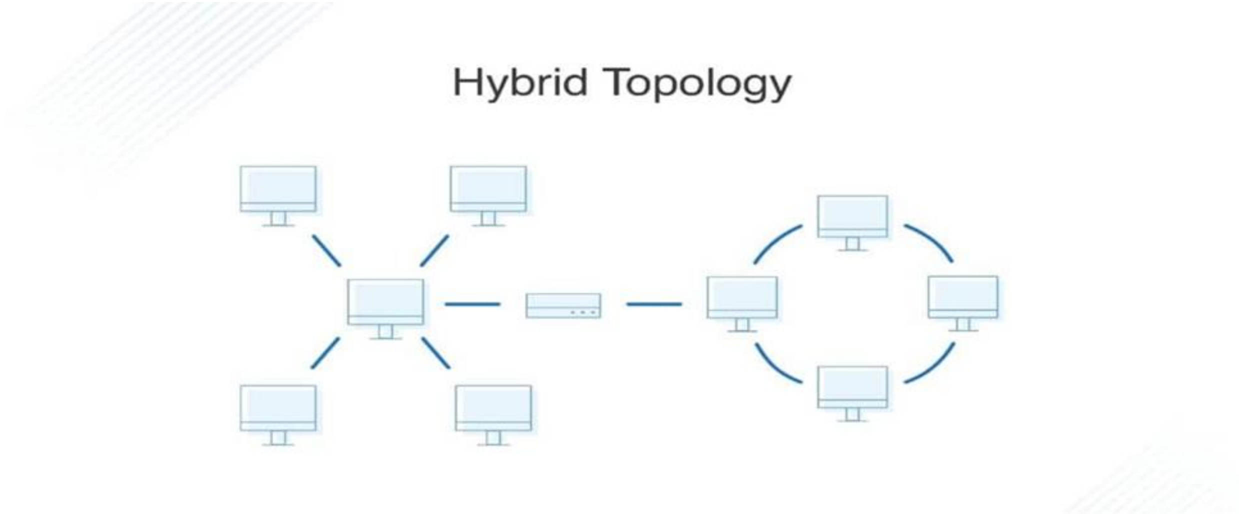
Disadvantages:

1. If a root node fails, the entire branch connected to it will stop working.
2. Tree topology requires more cables than simple topologies.
3. The design can be complex for large networks.
4. Maintenance and troubleshooting can be tricky due to the hierarchy.

6-Hybrid Topology

Hybrid Topology is the combination of all the various types of topologies we have studied above. Hybrid Topology is used when the nodes are free to take any form. It means these can be individuals such as Ring or Star topology or can be a combination of various types of topologies seen above. Each individual topology uses the protocol that has been discussed earlier.

Hybrid Topology




Hybrid topology is a network design that **combines two or more different topologies** to form a single network.

How it works (Flow):

1. Data travels within the local topology (such as star or ring).
2. It is then transferred to another topology through connecting devices.
3. The combined structure ensures efficient communication.
4. The data reaches the destination through the appropriate path.

Advantages:

1. Hybrid topology is flexible because it combines two or more topologies.
2. It is scalable and can be expanded easily.
3. It can take the benefits of multiple topologies at once.
4. If designed properly, it can provide high reliability.

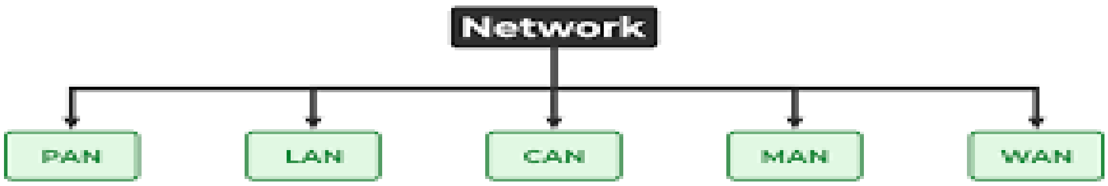
Disadvantages:

1. Hybrid topology is complex to design and implement.
2. It is more expensive than simple topologies.
3. Maintenance can be difficult due to the mixed structure.
4. Skilled management is required to operate it efficiently.

*Note: A common example of a hybrid topology is a university campus network. The network may have a backbone of a star topology, with each building connected to the backbone through a switch or router. Within each building, there may be a bus or ring topology connecting the different rooms and offices.*

Types of Networks:

PAN → LAN → CAN → MAN → WAN




## 1. PAN – Personal Area Network

**Definition**

A Personal Area Network (PAN) connects personal devices over a very short distance, usually around one person.

PAN is mainly used for **personal use**. It allows devices to communicate without cables and is common in daily life.

**Range**

Up to 10 meters (sometimes a little more with Wi-Fi)

**Diagram**

```
[Smartwatch]
      |
[Bluetooth]
      |
[Smartphone]---- [Earphones]
```

**Features**

- Very small coverage
- Low cost
- Mostly wireless (Bluetooth, Wi-Fi)

• Easy to set up

**Examples**

• Smartphone connected to Bluetooth earphones
• Smartwatch connected to a phone
• Laptop connected to a mobile hotspot

## 2. LAN – Local Area Network

**Definition**

A Local Area Network (LAN) connects computers and devices within a small area such as a home, office, or school.

LANs are commonly owned by individuals or organizations and are used to share files, printers, and internet connections.

**Range**

Up to 1 km (usually within a building)

**Diagram**

```
[PC]    [Laptop]
  \      /
   [Wi-Fi Router]
        |
     [Printer]
```

**Features**

• High speed
• Low error rate
• Easy to maintain
• Can be wired or wireless

Examples

• Home Wi-Fi network
• Office computer network
• School computer lab

## 3. CAN – Campus Area Network

**Definition**

A Campus Area Network (CAN) connects multiple LANs within a campus or organization.

CAN is used where many buildings need to stay connected under one organization, such as universities or hospitals.

**Range**

1 km to 5 km

**Diagram**

```
[Building A LAN]----\
                     \
[Building B LAN]------ [Campus Backbone]
                     /
[Building C LAN]----/
```

**Features**

• Larger than LAN, smaller than MAN

• High-speed backbone

• Centrally managed

**Examples**

• University campus network

• Hospital network connecting departments

• Corporate campus network

## 4. MAN – Metropolitan Area Network

**Definition**

A Metropolitan Area Network (MAN) covers a city or large town.

MAN connects many organizations across a city and is often used for public services and internet distribution.

**Range**

5 km to 50 km

**Diagram**

```
[Office LAN]----\
                 \
[College LAN]------ [City Network]
                 /
[Home LAN]-------/
```

**Features**

• Larger than CAN and LAN

• High-speed communication

• Usually owned by ISPs or government

Examples

• City-wide public Wi-Fi

• ISP network covering a city

• Government office networks in a city

## 5. WAN – Wide Area Network

Definition

A Wide Area Network (WAN) covers a very large geographical area, such as countries or continents.

WAN enables global communication. It is essential for banks, airlines, multinational companies, and the Internet.

Range

100 km to worldwide

Diagram

```
[Country A]---- Satellite/Fiber---- [Country B]
     \                    /
      ----------- Internet-------
```

Features

• Largest coverage

• Connects LANs, CANs, and MANs

• Slower than LAN due to distance

• Uses satellites, fiber optics, submarine cables

Examples

• The Internet (largest WAN)

• International banking networks

• Multinational company networks

| Network | Full Form | Range | Example |
|---|---|---|---|
| PAN | Personal Area Network | Up to 10 m | Bluetooth devices |
| LAN | Local Area Network | Up to 1 km | Home/School Wi-Fi |
| CAN | Campus Area Network | 1–5 km | University campus |
| MAN | Metropolitan Area Network | 5–50 km | City network |
| WAN | Wide Area Network | Worldwide | Internet |

## Internetworking

### Meaning of Internetworking

Internetworking is the process of connecting two or more computer networks so they can communicate and share resources.

### Example:

Connecting a school network (LAN) with the internet (WAN).

### Purpose of Internetworking

- To allow communication between different networks
- To share data, resources, and services
- To expand network coverage from local → global
- To improve reliability and scalability

### Major Internetworking Devices

1. Router
- Connects multiple networks
- Routes (selects best path) for data packets

2. Switch
- Connects multiple devices inside a LAN
- Works at data link layer

3. Hub
- Connects devices in a LAN but broadcasts data to all

4. Gateway
- Connects networks using different protocols

5. Modem
- Converts digital signals ↔ analog signals for internet access

### Characteristics of Internetworking

- Uses TCP/IP protocol.
- Enables data flow between heterogeneous networks.
- Increases communication range.

### Why Internetworking is Needed

- To share data
- To communicate
- To access the internet

- To connect networks over long distances

Applications Of Internet:-

1. Communication

- Email, WhatsApp, video calls
- Fast and easy way to talk to people anywhere

2. Information Sharing / Searching

- Google, Wikipedia, online articles
- Helps to find any information quickly

3. Social Networking

- Facebook, Instagram, X
- Used for connecting with friends and sharing updates

4. Online Shopping (E-commerce)

- Amazon, Flipkart
- Buying and selling products online

5. Online Education (E-learning)

- Online classes, YouTube learning, digital libraries
- Students can learn from home

6. Online Banking

- Money transfer, checking balance, bill payments
- No need to go to the bank physically

7. Entertainment

- Watching movies (Netflix), listening to music (Spotify), gaming
- Internet provides various forms of entertainment

8. E-Governance

- Online tax filing
- Applying for ID proofs and government services online

9. Cloud Storage

- Google Drive, OneDrive
- Saving files online and accessing them from anywhere

Network Models:

**Protocol:**

A protocol is a set of rules that define how data is transmitted between two devices on a network.

Protocols control format, order, error-checking, and speed of communication.

**Key features of a protocol**

- Syntax → Format of data
- Semantics → Meaning of data
- Timing → When data should be sent and how fast

**Protocols and Standards**

**Protocols**

A protocol is a set of rules that govern data communications. A protocol defines what is communicated, how it is communicated & when it is communicated. The key elements of a protocol are syntax, semantics & timing.

**a) Syntax:-**

The syntax defines the structure or format of data. This means that the order in which it is to be sent is decided. e.g. a protocol could define that first 8 bits of data to be the address of the sender, the second 8 bits to be the address of the receiver, & rest of the stream to be the message itself.

**b) Semantics:-**

Semantics define the interpretation of the data that is being sent.

e.g. if the last 2 bits of the receiver's address field contain 00, it means that the sender & the receiver are on the same network.

**c)Timing:-**

This refers to an agreement between the sender & the receiver about the data transmission rates & duration.

**Standards**

Standards provide guidelines to manufacturers, vendors, govt agencies, & other service providers to ensure the kind of interconnectivity necessary in today's market place & in international communications. Data communication standards fall into two categories.

a) *De facto:* - *De facto* standards are often established originally by manufacturers who seek to define the functionality of a new product or technology.
b) *De jure:* - These are the standards that have been legislated by an official body. They are usually led by governments or government appointed agencies.

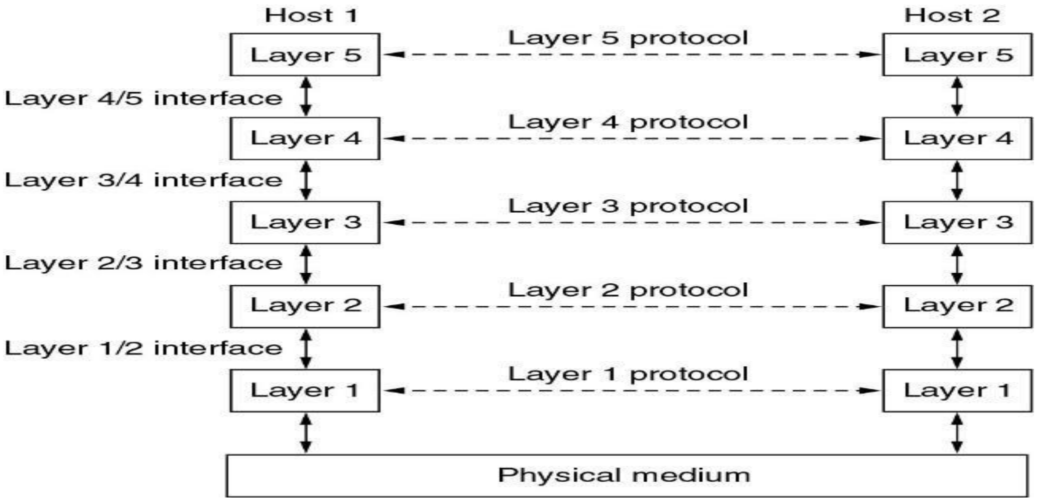
## Protocol Hierarchies

Most networks are organized as a stack of layers or levels, each one built upon the one below it. The number of layers, the name of each layer, the contents of each layer, and the function of each layer differ from network to network. The purpose of each layer is to offer certain services to the higher layers, shielding those layers from the details of how the offered services are actually implemented. In a sense, each layer is a kind of virtual machine, offering certain services to the layer above it.

Layer n on one machine carries on a conversation with layer n on another machine. The rules and conventions used in this conversation are collectively known as the layer n protocol. Basically, a protocol is an agreement between the communicating parties on how communication is to proceed.

A five-layer network is illustrated in Fig. The entities comprising the corresponding layers on different machines are called peers. The peers may be processes, hardware devices, or even human beings. In other words, it is the peers that communicate by using the protocol.

Figure . Layers, protocols, and interfaces.



In reality, no data are directly transferred from layer n on one machine to layer n on another machine. Instead, each layer passes data and control information to the layer immediately below it, until the lowest layer is reached. Below layer 1 is the physical medium through which actual communication occurs. In Fig. virtual communication is shown by dotted lines and physical communication by solid lines.

Between each pair of adjacent layers is an interface. The interface defines which primitive operations and services the lower layer makes available to the upper one.

A set of layers and protocols is called a network architecture. A list of protocols used by a certain system, one protocol per layer, is called a protocol stack.

## Design Issues for the Layers

Some of the key design issues that occur in computer networks are present in several layers. Below, we will briefly mention some of the more important ones.

### Addressing:-

Every layer needs a mechanism for identifying senders and receivers. Since a network normally has many computers, some of which have multiple processes, a means is needed for a process on one machine to specify with whom it wants to talk. As a consequence of having multiple destinations, some form of addressing is needed in order to specify a specific destination.

### Error control:-

It is an important issue because physical communication circuits are not perfect. Many error-detecting and error-correcting codes are known, but both ends of the connection must agree on which one is being used. In addition, the receiver must have some way of telling the sender which messages have been correctly received and which have not.

### Flow Control:-

Flow Control is needed at every level to keep a fast sender from swamping a slow receiver with data. Some kind of feedback is required from the receiver to the sender, either directly or indirectly, about the receiver's current situation. Others limit the sender to an agreed-on transmission rate. This is called as flow control.

**Multiplexing**:-
All processes are unable to accept arbitrarily long messages at several levels. This property leads to mechanisms for disassembling, transmitting, and then reassembling messages. A related issue is the problem of what to do when processes insist on transmitting data in units that are so small that sending each one separately is inefficient. Here the solution is to gather several small messages heading toward a common destination into a single large message and dismember the large message at the other side.

When it is inconvenient or expensive to set up a separate connection for each pair of communicating processes, the underlying layer may decide to use the same connection for multiple, unrelated conversations. As long as this multiplexing and demultiplexing is done transparently, it can be used by any layer. Multiplexing is needed in the physical layer, for example, where all the traffic for all connections has to be sent over at most a few physical circuits.

**Routing**:-

When there are multiple paths between source and destination, a route must be chosen. Sometimes this decision must be split over two or more layers. Then a low-level decision might have to made to select one of the available circuits based on the current traffic load. This topic is called routing.

**Connection-Oriented and Connectionless Services**
Connection-oriented and connectionless services are two types of communication services provided by computer networks.

**1)Connection-Oriented Services**:
In connection-oriented services, a connection is established between the sender and receiver before data transfer can occur. The connection is maintained throughout the duration of the communication, and data is transmitted in a continuous stream. Examples of connection-oriented services include the Transmission Control Protocol (TCP) used in the Internet, and Asynchronous Transfer Mode (ATM) networks.

1. **Example**: When you visit a website using a web browser, the browser establishes a TCP connection with the web server before requesting the web page. The connection is maintained until the web page has been fully downloaded, and then

it is closed. During the entire process, the data is transmitted over the same connection.

**2)Connectionless Services**:
In connectionless services, no prior connection is established between the sender and receiver before data transfer. Each data packet is treated as an independent unit, and there is no guarantee that the packets will arrive at the receiver in the same order they were sent. Examples of connectionless services include the User Datagram Protocol (UDP) used in the Internet, and the X.25 protocol used in wide-area networks.

**Example:**
When you send an email, the email message is broken down into packets, and each packet is sent independently to its destination. There is no guarantee that the packets will arrive at the destination in the same order they were sent, and some packets may be lost or corrupted during transmission. However, the packets are delivered as soon as they are received, without waiting for the other packets to arrive.
Overall, the choice between connection-oriented and connectionless services depends on the specific requirements of the application and the network. Connection-oriented services provide a reliable and ordered data transfer, while connectionless services provide a more efficient and flexible data transfer, with a lower overhead.

## Reference Models

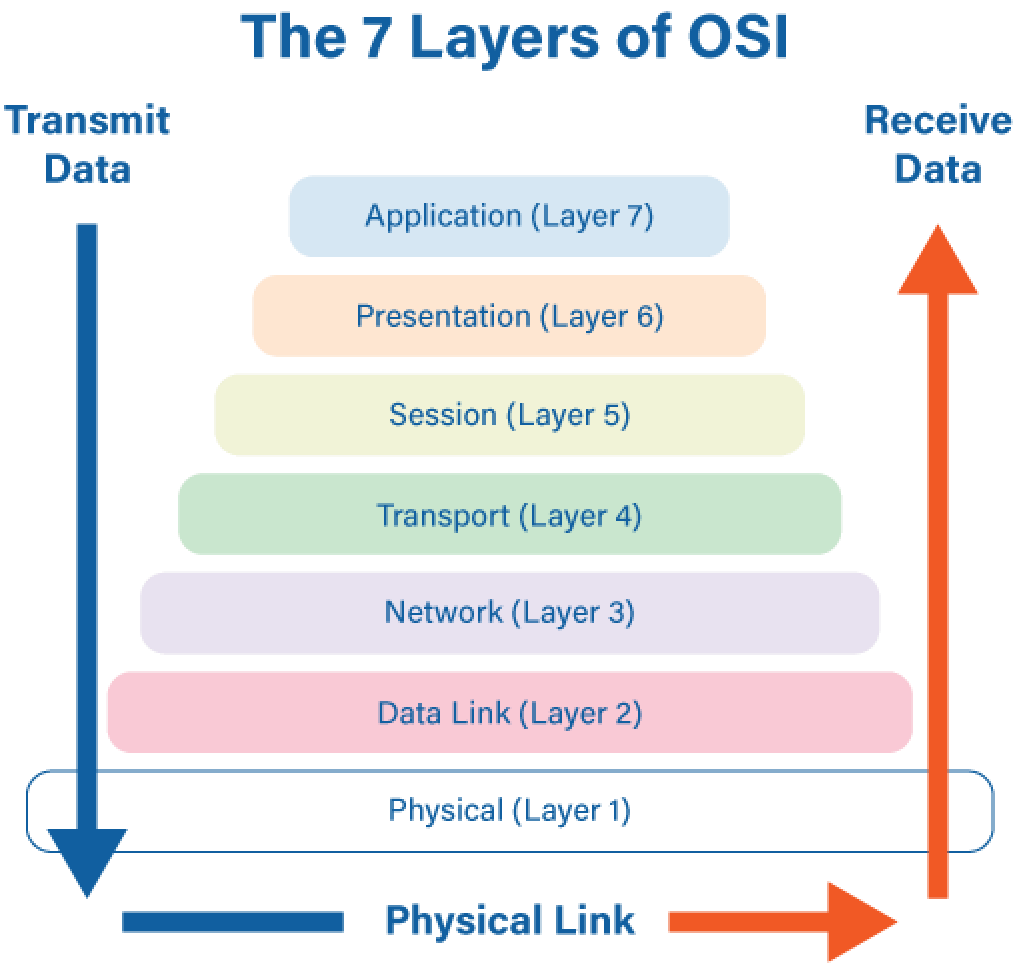
### OSI model (open system interconnection)

The purpose of the OSI model is to facilitate communication between different system without requiring changes to the logic of underlying hardware and software.

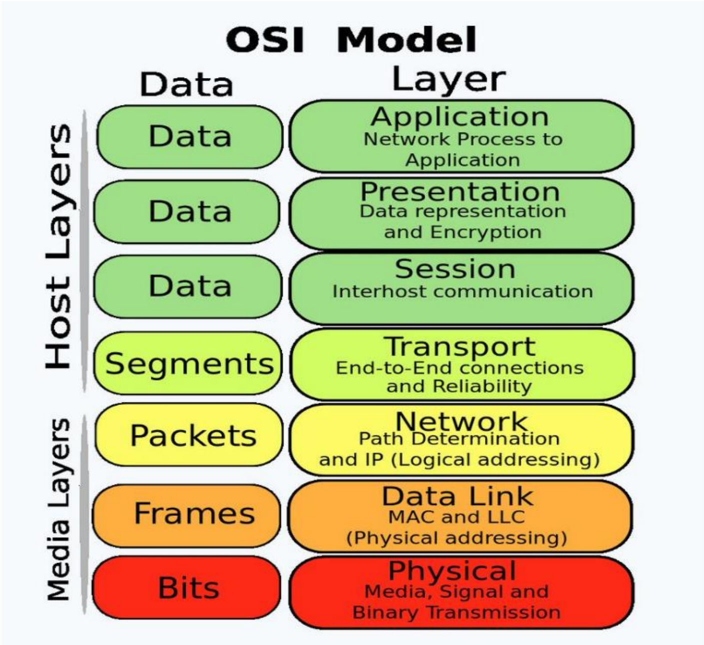
Now we will see how much layer in OSI model.

1. Application Layer
2. Presentation Layer
3. Session Layer
4. Transport Layer

5. Network Layer
6. Data link Layer
7. Physical Layer




| | | |
|---|---|---|
| 7 | Application Layer | Human-computer interaction layer, where applications can access the network services |
| 6 | Presentation Layer | Ensures that data is in a usable format and is where data encryption occurs |
| 5 | Session Layer | Maintains connections and is responsible for controlling ports and sessions |
| 4 | Transport Layer | Transmits data using transmission protocols including TCP and UDP |
| 3 | Network Layer | Decides which physical path the data will take |
| 2 | Data Link Layer | Defines the format of data on the network |
| 1 | Physical Layer | Transmits raw bit stream over the physical medium |




## OSI MODEL – 7 LAYERS (DETAILED)

### Physical Layer (Layer 1)

**Definition:** The Physical Layer is the lowest layer of the OSI model. It transmits raw bits. This layer defines hardware components. It handles physical communication.

**Purpose:** The purpose of this layer is to transmit bits. It defines cables and connectors. This layer manages voltage levels. It controls data rate.

**Examples:** Cables operate at this layer. Hubs work here. Repeaters function at this layer. Fiber optic transmission uses this layer.

**Working Flow:** Data is converted into bits. Bits are transmitted as signals. Signals travel through the medium. Data reaches the receiving device.

### Data Link Layer (Layer 2)

**Definition:** The Data Link Layer manages node-to-node delivery. It is the second layer of the OSI model. This layer handles physical addressing. It ensures error-free data transfer.

**Purpose:** The purpose of this layer is to frame data. It detects and corrects errors. This layer uses MAC addresses. It controls access to the physical medium.

**Examples:** Switches work at this layer. MAC addresses are used here. Ethernet operates at this layer. LAN communication uses this layer.

**Working Flow:** Packets are converted into frames. MAC addresses are added. Errors are checked using CRC. Frames are sent to the Physical Layer.

### Network Layer (Layer 3)

**Definition:** The Network Layer handles routing of data. It is the third layer of the OSI model. This layer manages logical addressing. It selects the best path for data transfer.

**Purpose:** The purpose of this layer is to route data correctly. It manages IP addressing. This layer ensures packet delivery across networks. It handles congestion control.

**Examples:** Routers operate at this layer. IP addressing is handled here. ICMP messages work here. Routing protocols function at this layer.

**Working Flow:** Data is converted into packets. The best path is determined. Packets are forwarded to the next network. Data is sent to the Data Link Layer.

### Transport Layer (Layer 4)

**Definition:** The Transport Layer ensures reliable data delivery. It is the fourth layer of the OSI model. This layer divides data into segments. It controls end-to-end communication.

**Purpose:** The purpose of the Transport Layer is error-free data delivery. It controls flow of data between sender and receiver. This layer manages port numbers. It ensures correct sequencing of data.

**Examples:** TCP protocol works at this layer. UDP protocol also operates here. File downloads use TCP. Video streaming often uses UDP.

**Working Flow:** Data from upper layers is segmented here. Each segment is numbered for order. Errors are checked and corrected. Data is then sent to the Network Layer.

### Session Layer (Layer 5)

**Definition:** The Session Layer manages communication sessions between devices. It is the fifth layer of the OSI model. This layer controls the session connection. It keeps communication organized.

**Purpose:** The purpose of this layer is to establish and maintain sessions. It manages dialog control between devices. This layer provides synchronization points. It helps recover sessions after interruption.

**Examples:** Video call sessions use the Session Layer. Online banking login sessions work here. User authentication sessions depend on this layer. Chat applications also use this layer.

**Working Flow:** The Session Layer establishes a connection between devices. It keeps the session active during communication. It monitors the data exchange. Finally, it terminates the session properly.

### Presentation Layer (Layer 6)

**Definition:** The Presentation Layer is the sixth layer of the OSI model. It prepares data before transmission. This layer ensures data is readable by the receiver. It acts as a translator between systems.

**Purpose:** The purpose of the Presentation Layer is to format data properly. It provides data encryption and decryption for security. This layer also compresses data to reduce size. It ensures compatibility between devices.

**Examples:** Data encryption using HTTPS is an example of this layer. File compression before sending data works here. Image and audio formatting are handled at this layer. Secure data transfer uses this layer.

**Working Flow:** Data received from the Application Layer is formatted here. The layer encrypts the data if required. It compresses the data for faster transfer. Then it sends data to the Session Layer.

### Application Layer (Layer 7)

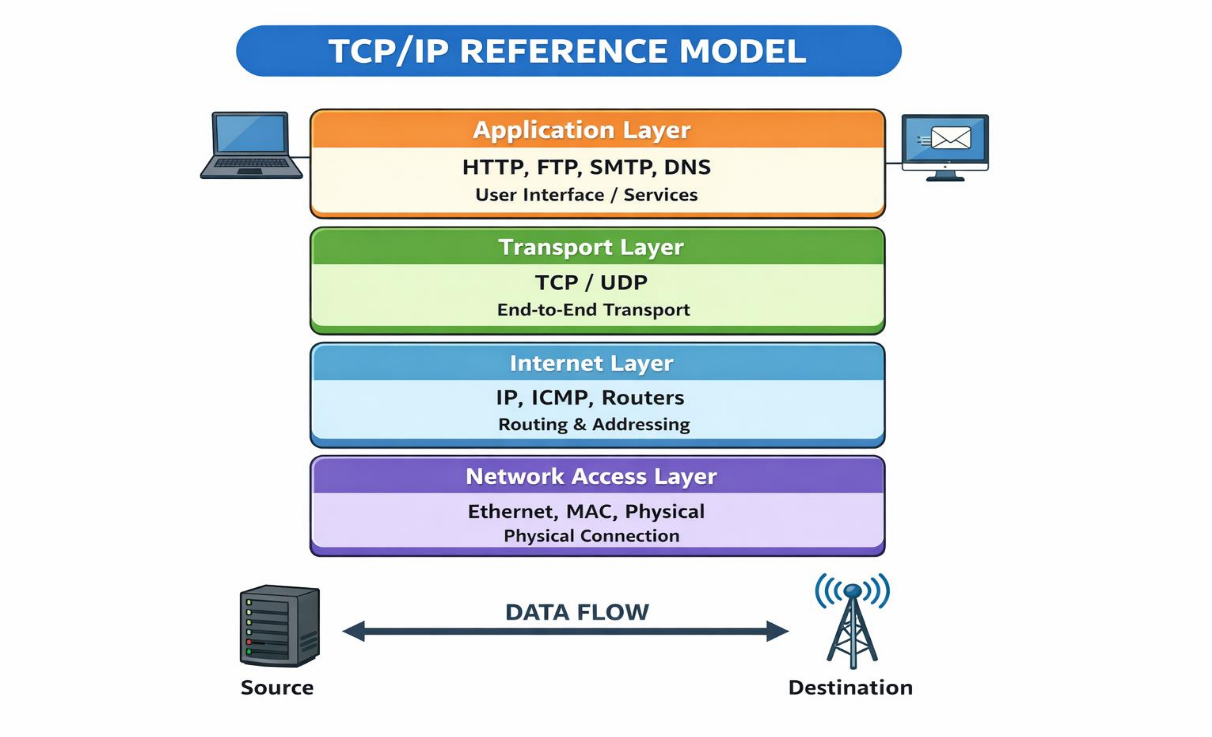
**Definition:** The Application Layer is the topmost layer of the OSI model. It provides network services directly to the user. This layer helps users communicate with the network easily. It is responsible for starting network communication.

**Purpose:** The main purpose of the Application Layer is to allow user applications to access network services. It helps in sending and receiving data over the network. This layer supports services such as email, file transfer, and web browsing. It ensures proper interaction between user software and the network.

**Examples:** Opening a website using a web browser is an example of the Application Layer. Sending an email through Gmail also works at this layer. Downloading or uploading files using FTP is handled here. Remote login services like Telnet are other common examples.

**Working Flow:** When a user performs an action such as opening a website, the request is sent to the Application Layer. This layer receives the request from the user application. It selects the appropriate protocol like HTTP or FTP. Then it forwards the data to the Presentation Layer for further processing.

## TCP/IP reference model.




TCP/IP Reference Model

The **TCP/IP (Transmission Control Protocol / Internet Protocol) reference model** is a practical networking model used for real-world communication on the Internet. It has **4 layers**, and it explains how data is sent from one device to another.

1)Application Layer

**Definition:** The Application Layer in the TCP/IP model provides network services directly to the user. It combines the functions of the OSI Application, Presentation, and Session layers. This layer allows user applications to communicate over the network. It is the top layer of the TCP/IP model.

**Purpose:** The main purpose of this layer is to allow users to access network services. It supports communication services like web browsing and email. This layer helps applications send and receive data. It ensures proper interaction between user software and the network.

**Examples:** Opening a website using a browser is an example of this layer. Sending or receiving emails works at this layer. File transfer using FTP is handled here. Protocols like HTTP, FTP, and SMTP are used in this layer.

**Working Flow:** The user starts an action such as opening a website. The Application Layer receives this request from the user application. It prepares the data using the correct protocol. Then it sends the data to the Transport Layer.

### 2)Transport Layer

**Definition:** The Transport Layer is responsible for end-to-end communication between devices. It ensures data is delivered from sender to receiver. This layer divides data into smaller units. It controls reliable or fast transmission.

**Purpose:** The purpose of this layer is to provide reliable data transfer. It controls errors and data flow. This layer uses port numbers to identify applications. It ensures data reaches the correct application.

**Examples:** TCP protocol works at this layer for reliable communication. UDP protocol also works here for faster communication. File downloads use TCP. Video streaming often uses UDP.

**Working Flow:** Data is received from the Application Layer. The Transport Layer divides data into segments. It adds port numbers and error control information. Then it sends the data to the Internet Layer.

### 3)Internet Layer

**Definition:** The Internet Layer is responsible for logical addressing and routing. It decides how data packets move across networks. This layer is similar to the OSI Network Layer. It uses IP addresses for communication.

**Purpose:** The purpose of this layer is to route data from source to destination. It selects the best path for data transfer. This layer ensures packets reach the correct network. It manages packet forwarding.

**Examples:** IP protocol works at this layer. Routers operate at this layer. ICMP protocol is also used here. IP addressing is handled in this layer.

**Working Flow:** Data is received from the Transport Layer. The Internet Layer adds IP address information. It converts data into packets. Then it forwards the packets to the Network Access Layer.

### 4)Network Access Layer

**Definition:** The Network Access Layer is the lowest layer of the TCP/IP model. It handles physical transmission of data. This layer includes hardware and physical addressing. It combines OSI Data Link and Physical layers.

**Purpose:** The purpose of this layer is to transmit data over the physical network. It controls hardware communication. This layer manages MAC addresses. It ensures data is sent as signals.

**Examples:** Ethernet works at this layer. Network cables and switches are used here. MAC addresses are handled here. LAN communication depends on this layer.

**Working Flow:** Packets are received from the Internet Layer. They are converted into frames and bits. Data is transmitted as electrical or optical signals. The data reaches the destination device.

## Conclusion

The TCP/IP reference model is widely used for Internet communication. It is simpler than the OSI model and more practical. Each layer has a specific role in data transmission. Together, all layers ensure successful network communication.